

Depreciation Coverage Moderation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Depreciation Coverage Moderation (DCM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCM achieves an annualized gross (net) Sharpe ratio of 0.42 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (12) bps/month with a t-statistic of 2.44 (1.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets) is 14 bps/month with a t-statistic of 2.01.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding the cross-section of returns, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a novel predictor of stock returns based on firms' depreciation coverage patterns that captures time-varying exposure to aggregate consumption dynamics. Depreciation Coverage Moderation (DCM) measures the stability of a firm's ability to cover depreciation expenses through operational cash flows, reflecting the firm's resilience to consumption shocks and its covariance with the stochastic discount factor.

Firms with moderate depreciation coverage exhibit distinct exposure to consumption risk through their operational flexibility and investment irreversibility. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with extreme depreciation coverage ratios face heightened sensitivity to persistent shocks in consumption growth, as their capital allocation decisions become increasingly rigid during periods of consumption uncertainty. The consumption-based model with habit formation ([Campbell and Cochrane, 1999](#)) predicts that firms maintaining moderate depreciation coverage provide insurance against adverse consumption states, as their operational flexibility allows them to adjust capital intensity when the surplus consumption ratio declines. During economic downturns characterized by elevated marginal utility, firms with moderate depreciation coverage can reallocate resources more efficiently, reducing their covariance with the pricing kernel ([Cochrane et al., 2008](#)). The disaster risk model of [Barro \(2006\)](#) further implies that depreciation coverage moderation signals lower exposure to rare consumption disasters. Firms with balanced depreciation coverage maintain operational buffers that prove valuable when disaster risk materializes, as their capital structure allows for rapid adjustment without trig-

gering financial constraints. This state-dependent risk exposure manifests through the intertemporal marginal rate of substitution, where investors demand higher returns from firms whose cash flows covary strongly with consumption growth ([Parker and Julliard, 2005](#)).

The DCM trading strategy delivers economically and statistically significant returns across multiple specifications. A value-weighted long/short portfolio based on DCM achieves an annualized gross Sharpe ratio of 0.42, with monthly average excess returns of 21 basis points (t-statistic = 3.04). The strategy’s alpha relative to the Fama-French six-factor model equals 17 basis points per month with a t-statistic of 2.44, demonstrating robustness to standard risk adjustments. The economic magnitude remains substantial after accounting for transaction costs, with the net Sharpe ratio reaching 0.31 and the net six-factor alpha equaling 12 basis points per month (t-statistic = 1.67). Among the largest stocks exceeding the 80th NYSE percentile, the DCM strategy generates average returns of 32 basis points per month (t-statistic = 3.51), with six-factor alphas of 26 basis points (t-statistic = 2.83). The strategy’s performance persists across alternative portfolio constructions, with nineteen of twenty-five specifications producing t-statistics exceeding three for gross alphas. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the DCM strategy maintains an alpha of 14 basis points per month with a t-statistic of 2.01. The strategy’s gross Sharpe ratio exceeds 83% of documented anomalies in the factor zoo, while its net Sharpe ratio surpasses 91% of existing strategies after transaction costs.

This paper extends the consumption-based asset pricing literature by identifying a firm characteristic that captures cross-sectional variation in consumption risk exposure through operational flexibility. While [Zhang \(2005\)](#) document the role of investment irreversibility in expected returns and [Cooper et al. \(2008\)](#) establish the asset growth anomaly, we demonstrate that depreciation coverage moderation pro-

vides distinct information about firms’ state-dependent risk profiles beyond existing investment-based predictors. Our findings complement [Lettau and Ludvigson \(2001\)](#) by showing how firm-level operational characteristics relate to aggregate consumption dynamics, with DCM capturing variation in the cross-section that traditional consumption-based factors miss. The methodological innovation lies in constructing a measure that simultaneously captures firms’ exposure to long-run consumption risks and their resilience to disaster states, addressing the challenge raised by [Savov \(2011\)](#) regarding the measurement of consumption-based risk in financial markets. Unlike [Yogo \(2006\)](#), who focuses on durable consumption growth as a risk factor, we identify a firm characteristic that predicts returns through its correlation with the broader consumption process, including both normal and disaster states. The DCM signal improves the investment opportunity set even after controlling for established anomalies, with the augmented combination strategy growing from

993.77to

1194.95, demonstrating its incremental contribution to the factor zoo. These results provide new evidence supporting consumption-based models while offering practical insights for portfolio construction in the presence of time-varying consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and depreciation expenses across firms. construction of our Depreciation Coverage Moderation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents

the cash proceeds from the issuance of long-term debt obligations during the fiscal year, capturing firms' external financing activities through debt markets. Depreciation, depletion, and amortization (DPACT) measures the total non-cash charges against earnings that reflect the allocation of costs associated with tangible and intangible assets over their useful lives, as well as the depletion of natural resources. This variable represents the periodic expense recognition for capital investments and acquired assets. We construct the Depreciation Coverage Moderation signal by calculating the year-over-year change in long-term debt issuance scaled by lagged depreciation expense, specifically computing $(DLTIS(t) - DLTIS(t-1)) / DPACT(t-1)$. This ratio captures the change in a firm's debt financing activity relative to its depreciation base, providing insight into how firms moderate their external financing needs in relation to their non-cash expense recognition. The signal effectively measures whether firms are accelerating or decelerating their debt issuance relative to their depreciation expense levels, which may reflect changes in investment opportunities, capital structure adjustments, or financing constraints. We calculate this measure using fiscal year-end values and require non-missing observations for both DLTIS and DPACT in consecutive years. To ensure meaningful ratios, we exclude firm-year observations where lagged DPACT equals zero or is negative, which would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCM signal. Panel A plots the time-series of the mean, median, and interquartile range for DCM. On average, the cross-sectional mean (median) DCM is -2.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCM data. The signal's interquartile range spans -0.42 to 0.44. Panel B of Figure 1 plots the time-

series of the coverage of the DCM signal for the CRSP universe. On average, the DCM signal is available for 5.82% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does DCM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCM portfolio and sells the low DCM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DCM strategy earns an average return of 0.21% per month with a t-statistic of 3.04. The annualized Sharpe ratio of the strategy is 0.42. The alphas range from 0.17% to 0.26% per month and have t-statistics exceeding 2.44 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 5.24 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 473 stocks and an average market capitalization of at least \$1,672 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.74. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -9-27bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.45. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in

eighteen cases.

Table 3 provides direct tests for the role size plays in the DCM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCM, as well as average returns and alphas for long/short trading DCM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCM strategy achieves an average return of 32 bps/month with a t-statistic of 3.51. Among these large cap stocks, the alphas for the DCM strategy relative to the five most common factor models range from 26 to 35 bps/month with t-statistics between 2.83 and 3.92.

5 How does DCM perform relative to the zoo?

Figure 2 puts the performance of DCM in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCM strategy falls in the distribution. The DCM strategy’s gross (net) Sharpe ratio of 0.42 (0.31) is greater than 83% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCM strategy (red line).² Ignoring trading costs, a \$1 invested in the DCM strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.21 which ranks the DCM strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCM strategy would have yielded \$1.25 which ranks the DCM strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCM relative to those. Panel A shows that the DCM strategy gross alphas fall between the 51 and 62 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCM strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCM ranks between the 66 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCM with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCM or at least to weaken the power DCM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCM signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.91, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DCM is 1.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCM strategy earns alphas that range from 14-16bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCM trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.01.

7 Does DCM add relative to the whole zoo?

Finally, we can ask how much adding DCM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DCM signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCM grows to \$1194.95.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCM is available.

8 Conclusion

This study provides compelling evidence that Depreciation Coverage Moderation (DCM) represents a meaningful and robust signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCM possesses economically significant predictive power that persists even after accounting for well-established risk factors and related signals from the factor zoo.

The key findings of our research reveal that a value-weighted long/short trading strategy based on DCM generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.42 and a net Sharpe ratio of 0.31 after accounting for transaction costs. The signal’s robustness is particularly noteworthy, as it delivers statistically significant abnormal returns of 17 basis points per month (t-statistic = 2.44) relative to the Fama-French five-factor model augmented with momentum. Even after controlling for six closely related strategies including net debt financing, changes in financial liabilities, and various growth measures, DCM maintains a significant alpha of 14 basis points per month (t-statistic = 2.01), suggesting that it captures unique information content not subsumed by existing factors.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The persistence of DCM’s predictive power after transaction costs indicates its potential viability as an implementable investment strategy. The signal appears to capture a distinct aspect of firm fundamentals related to the balance between depreciation policies and coverage ratios that is not fully reflected in traditional value, profitability, or investment factors. This suggests that investors may benefit from incorporating DCM into their factor models and stock selection processes.

However, several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample

period and universe of stocks examined may influence the generalizability of our findings. Second, the economic mechanism underlying DCM's predictive power requires further theoretical development and empirical investigation. Third, the strategy's performance during different market regimes and its sensitivity to macroeconomic conditions remain unexplored.

Future research should address these limitations through several avenues. First, examining DCM's performance in international markets would provide valuable out-of-sample evidence regarding its robustness across different institutional and accounting environments. Second, investigating the time-varying nature of DCM's predictive power and its relationship with business cycles could enhance our understanding of when the signal is most effective. Third, exploring the fundamental economic channels through which depreciation coverage moderation affects firm value and investor expectations would strengthen the theoretical foundation of this anomaly. Finally, examining potential interactions between DCM and other accounting-based signals may reveal complementarities that could further improve return prediction. Such investigations would not only validate the robustness of our findings but also contribute to a more comprehensive understanding of how depreciation and coverage metrics influence asset prices in financial markets.

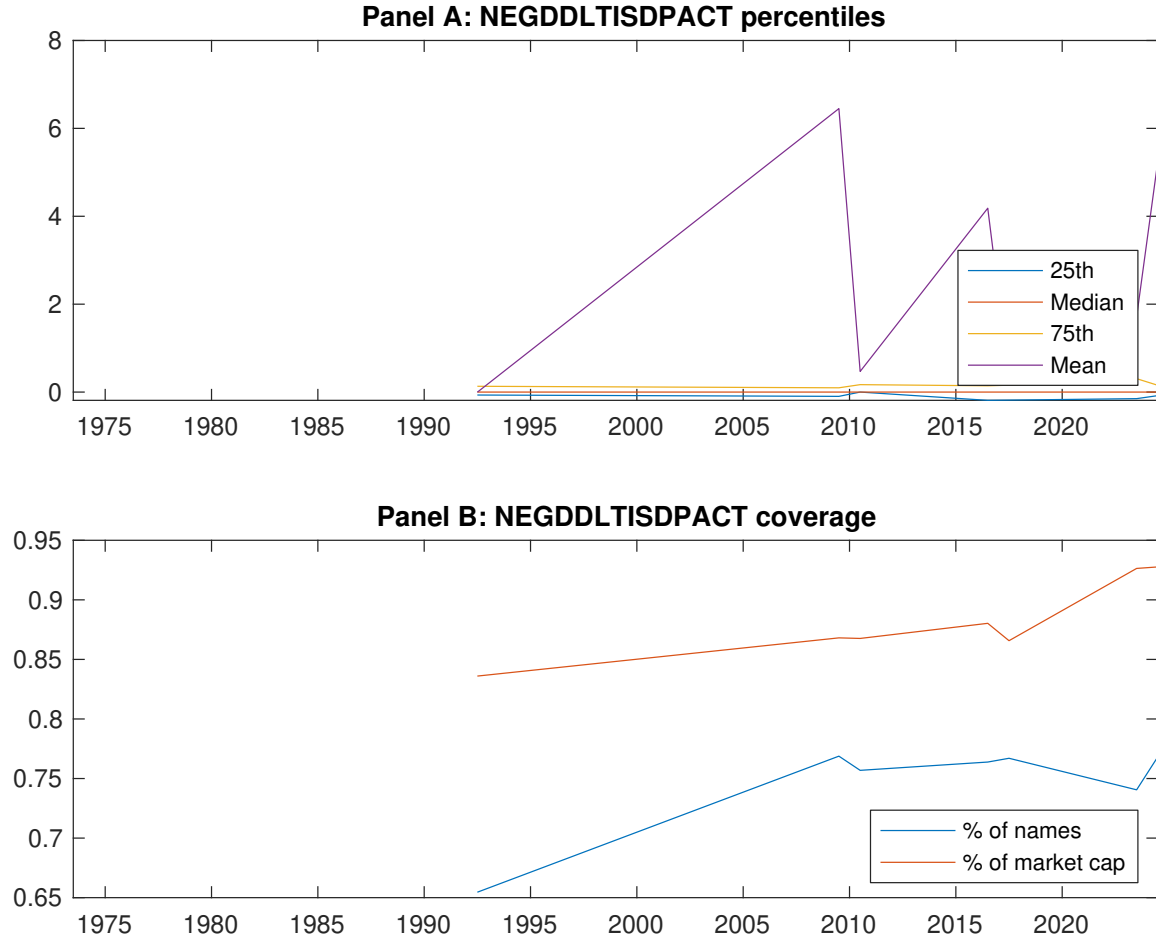


Figure 1: Times series of DCM percentiles and coverage.
This figure plots descriptive statistics for DCM. Panel A shows cross-sectional percentiles of DCM over the sample. Panel B plots the monthly coverage of DCM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.66 [3.00]	0.63 [3.63]	0.65 [3.31]	0.69 [4.00]	0.87 [4.25]	0.21 [3.04]
α_{CAPM}	-0.09 [-1.70]	0.05 [0.92]	-0.01 [-0.16]	0.11 [2.17]	0.17 [3.22]	0.26 [3.83]
α_{FF3}	-0.08 [-1.49]	0.02 [0.48]	0.05 [0.96]	0.11 [2.18]	0.19 [3.59]	0.26 [3.82]
α_{FF4}	-0.06 [-1.08]	0.03 [0.55]	0.10 [1.72]	0.08 [1.53]	0.17 [3.24]	0.23 [3.26]
α_{FF5}	-0.05 [-0.99]	-0.06 [-1.21]	0.08 [1.47]	0.02 [0.40]	0.14 [2.65]	0.19 [2.75]
α_{FF6}	-0.04 [-0.75]	-0.05 [-1.02]	0.12 [2.04]	0.00 [0.06]	0.13 [2.48]	0.17 [2.44]
Panel B: Fama and French (2018) 6-factor model loadings for DCM-sorted portfolios						
β_{MKT}	1.11 [89.00]	0.95 [83.22]	0.97 [73.04]	0.94 [82.99]	1.05 [85.28]	-0.06 [-3.40]
β_{SMB}	0.09 [4.99]	-0.12 [-6.87]	-0.01 [-0.67]	-0.05 [-3.18]	0.15 [8.10]	0.06 [2.32]
β_{HML}	-0.01 [-0.24]	0.06 [2.67]	-0.17 [-6.55]	-0.07 [-3.31]	-0.12 [-5.05]	-0.11 [-3.63]
β_{RMW}	0.03 [1.34]	0.16 [7.13]	-0.03 [-1.06]	0.08 [3.66]	0.06 [2.59]	0.03 [0.94]
β_{CMA}	-0.14 [-3.94]	0.10 [3.04]	-0.05 [-1.36]	0.23 [6.96]	0.11 [2.96]	0.25 [5.24]
β_{UMD}	-0.02 [-1.61]	-0.01 [-1.24]	-0.05 [-3.90]	0.03 [2.31]	0.01 [1.01]	0.03 [1.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	473	989	520	645	
me (\$10 ⁶)	1700	2544	2113	2579	1672	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.04]	0.26 [3.83]	0.26 [3.82]	0.23 [3.26]	0.19 [2.75]	0.17 [2.44]
Quintile	NYSE	EW	0.17 [3.74]	0.19 [4.34]	0.18 [4.07]	0.18 [3.83]	0.17 [3.69]	0.17 [3.60]
Quintile	Name	VW	0.23 [3.19]	0.28 [3.82]	0.28 [3.80]	0.25 [3.34]	0.23 [3.10]	0.21 [2.85]
Quintile	Cap	VW	0.23 [3.54]	0.27 [4.32]	0.27 [4.24]	0.24 [3.64]	0.20 [3.08]	0.18 [2.75]
Decile	NYSE	VW	0.34 [3.65]	0.40 [4.30]	0.37 [4.04]	0.33 [3.51]	0.26 [2.78]	0.23 [2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.19]	0.20 [2.85]	0.20 [2.81]	0.17 [2.52]	0.13 [1.91]	0.12 [1.67]
Quintile	NYSE	EW	-0.09 [-1.45]					
Quintile	Name	VW	0.17 [2.35]	0.22 [2.99]	0.22 [2.95]	0.20 [2.72]	0.18 [2.39]	0.16 [2.20]
Quintile	Cap	VW	0.18 [2.72]	0.23 [3.57]	0.22 [3.47]	0.20 [3.16]	0.16 [2.50]	0.14 [2.27]
Decile	NYSE	VW	0.27 [2.87]	0.33 [3.50]	0.30 [3.25]	0.28 [2.99]	0.20 [2.20]	0.19 [2.02]

Table 3: Conditional sort on size and DCM

This table presents results for conditional double sorts on size and DCM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCM and short stocks with low DCM. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCM Quintiles					DCM Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.64 [2.19]	0.93 [3.44]	0.95 [3.40]	1.06 [3.60]	0.74 [2.56]	0.10 [1.25]	0.13 [1.60]	0.11 [1.29]	0.08 [0.94]	0.11 [1.35]	0.09 [1.11]
	(2)	0.77 [2.79]	0.88 [3.57]	0.82 [3.26]	0.92 [3.76]	0.85 [3.30]	0.08 [1.04]	0.12 [1.55]	0.09 [1.16]	0.11 [1.43]	0.06 [0.77]	0.08 [1.03]
	(3)	0.79 [3.10]	0.82 [3.74]	0.80 [3.35]	0.84 [3.80]	0.93 [3.82]	0.14 [1.75]	0.17 [2.14]	0.17 [2.05]	0.15 [1.77]	0.17 [2.00]	0.15 [1.80]
	(4)	0.73 [3.15]	0.82 [3.96]	0.84 [3.77]	0.79 [3.82]	0.85 [3.86]	0.12 [1.64]	0.16 [2.11]	0.13 [1.76]	0.10 [1.32]	0.10 [1.30]	0.08 [1.01]
	(5)	0.55 [2.71]	0.55 [3.19]	0.61 [3.24]	0.60 [3.35]	0.87 [4.47]	0.32 [3.51]	0.35 [3.84]	0.35 [3.92]	0.30 [3.28]	0.30 [3.24]	0.26 [2.83]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCM Quintiles					DCM Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	366	368	368	368	363	33	30	29	29	32	
	(2)	101	102	101	102	101	55	56	55	55	55	
	(3)	72	73	72	73	72	100	99	96	98	99	
	(4)	61	62	62	62	61	218	226	218	223	215	
(5)	57	57	57	57	57	1451	1936	1684	2047	1467		

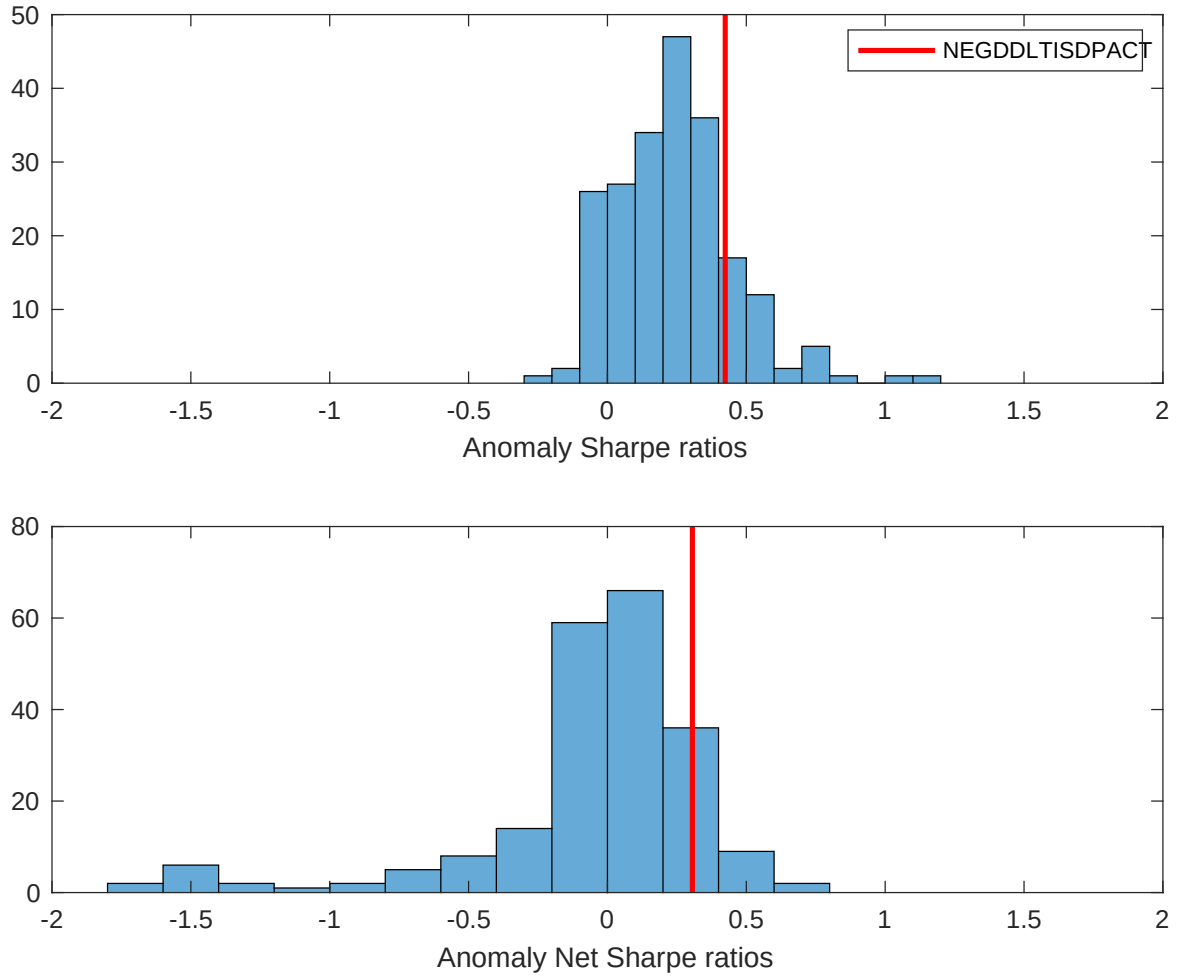


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

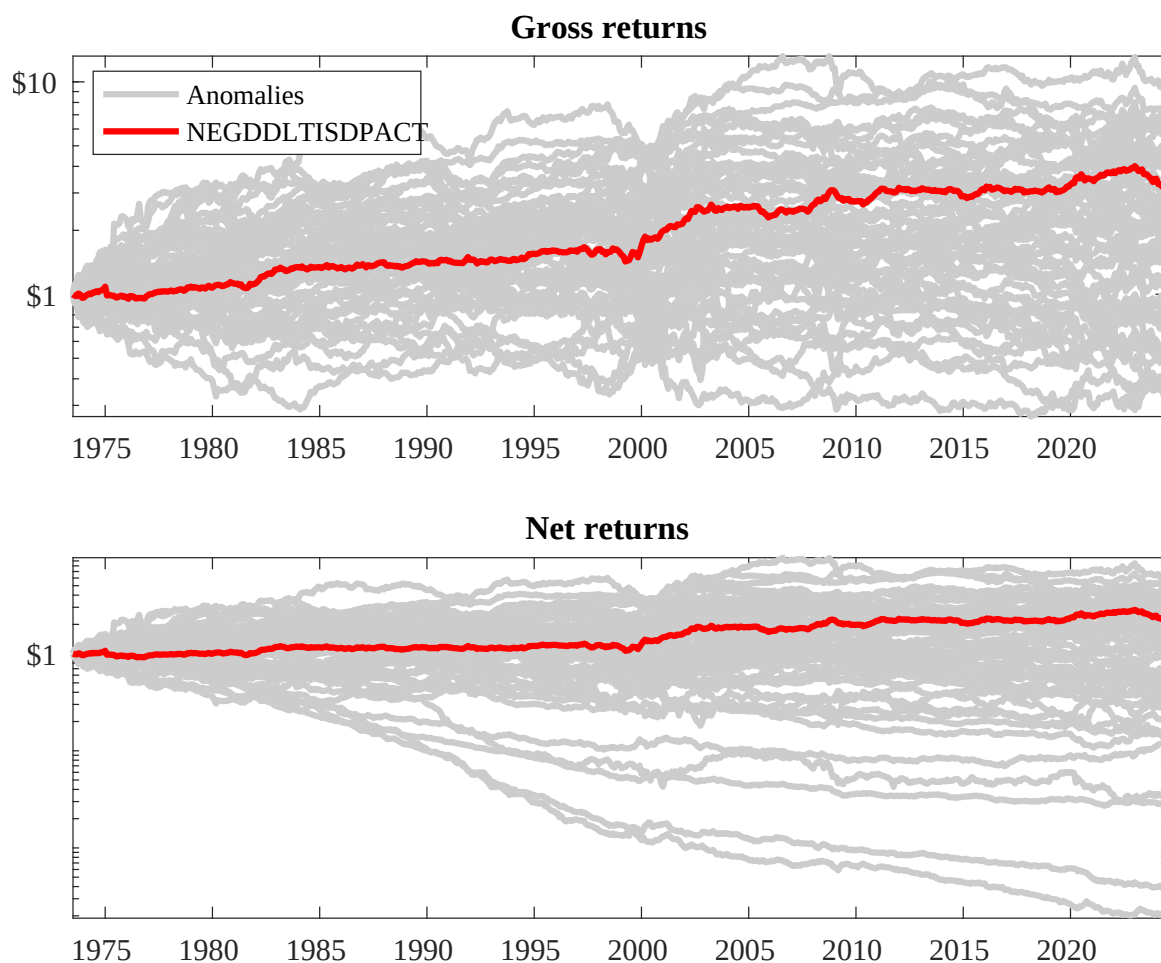
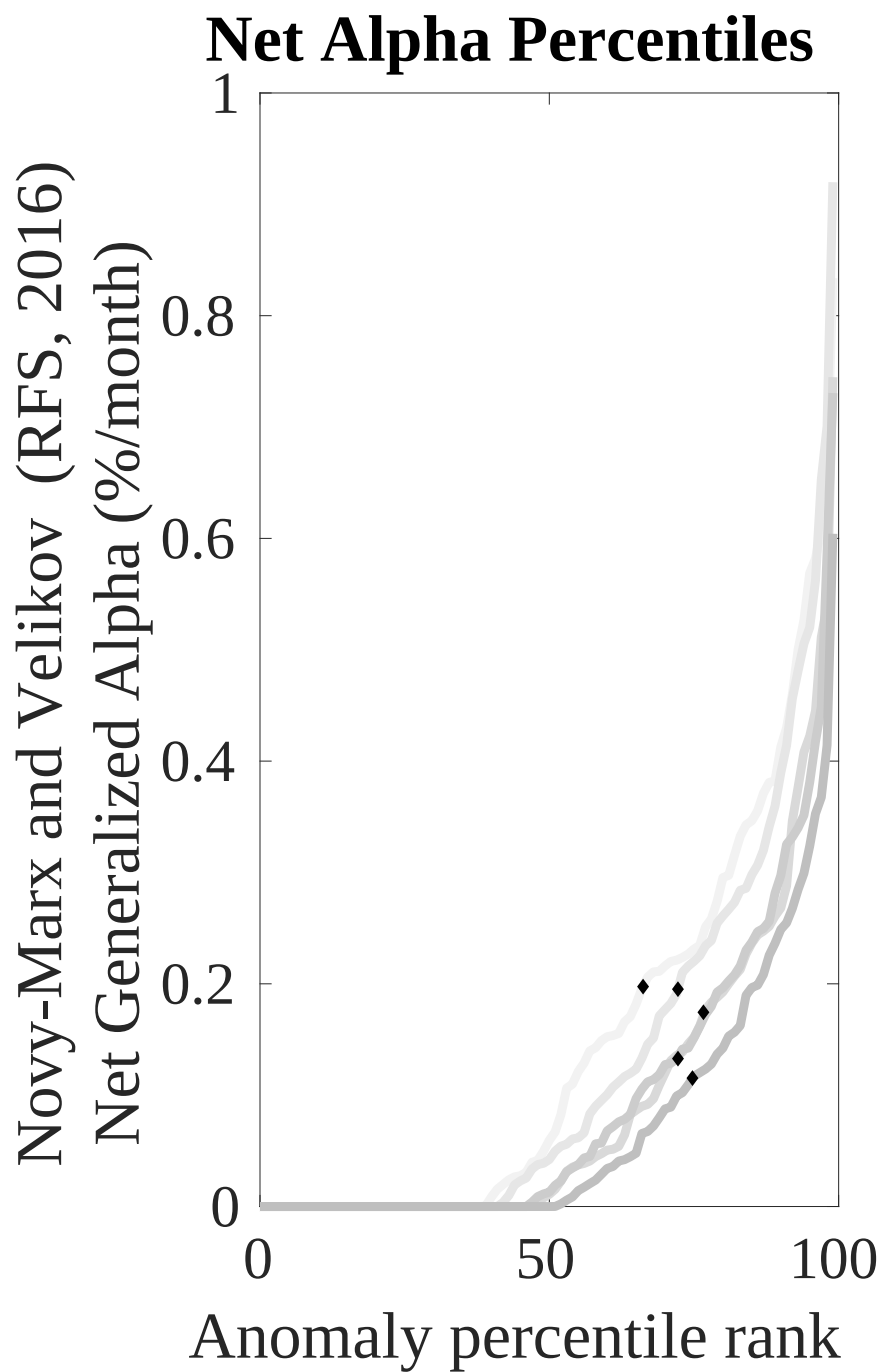
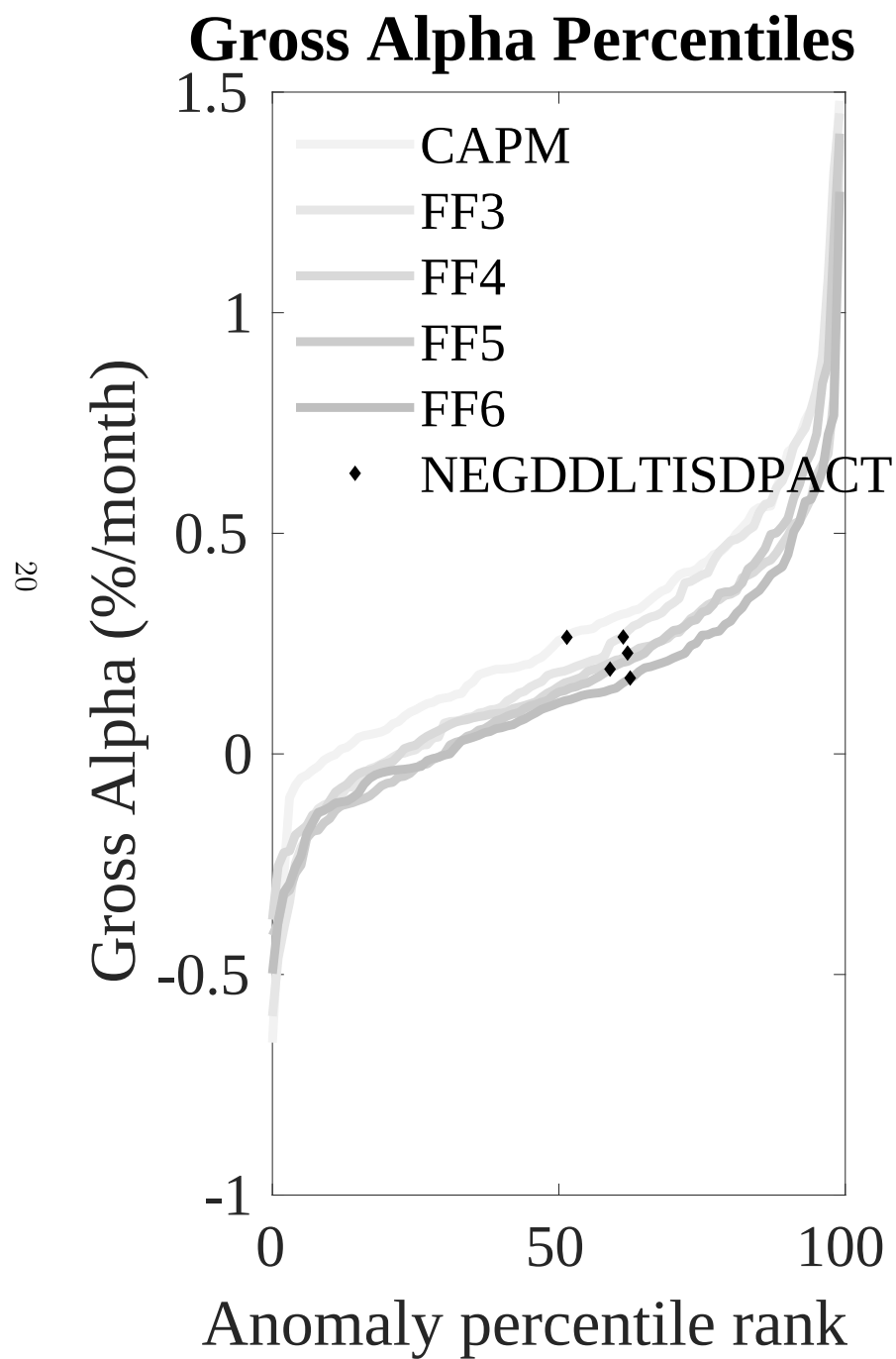


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



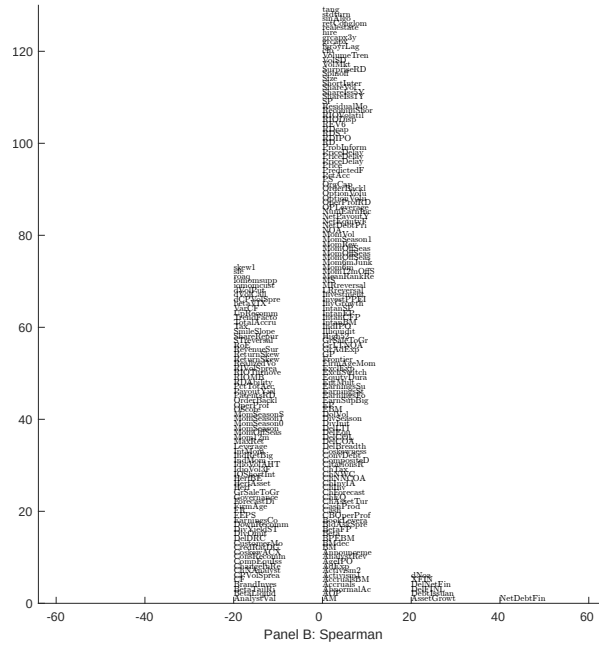
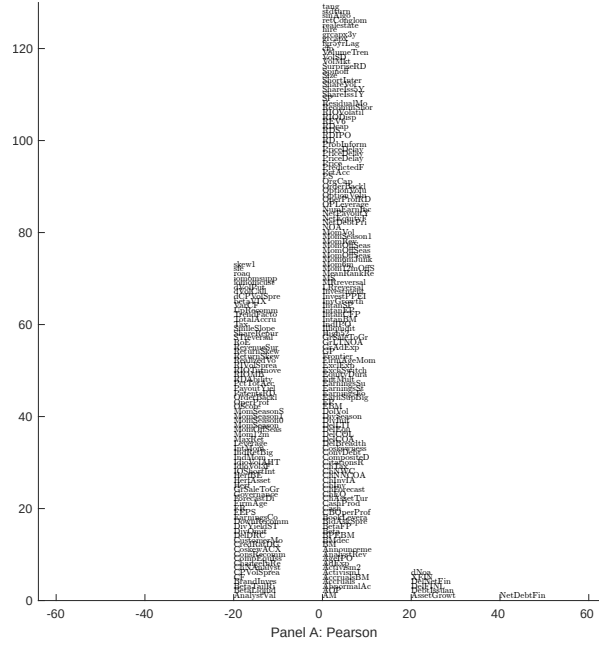


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with DCM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

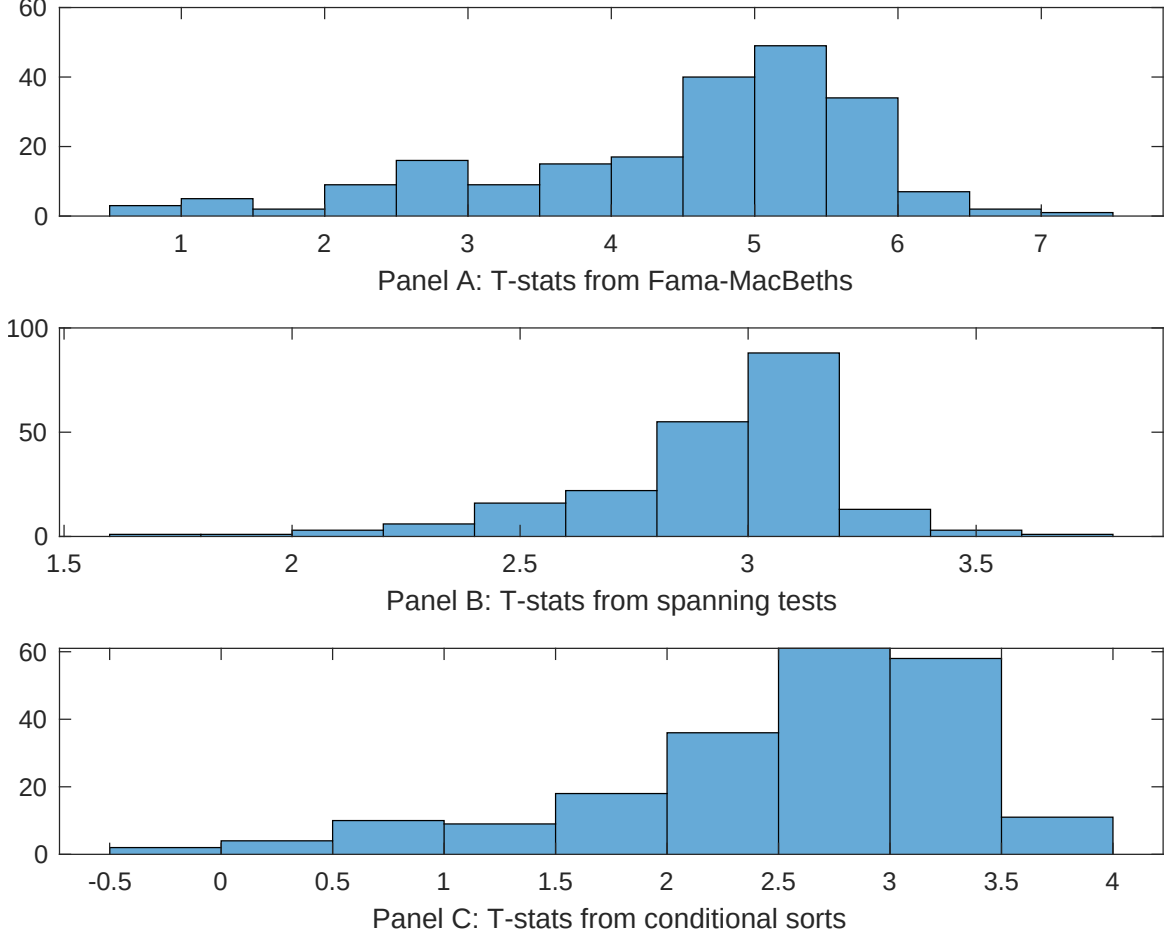


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCM conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCM. Stocks are finally grouped into five DCM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCM trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCM}DCM_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.35]	0.13 [5.38]	0.14 [5.72]	0.14 [5.30]	0.14 [5.79]	0.14 [5.76]	0.15 [5.74]
DCM	0.21 [2.65]	0.19 [2.78]	0.24 [3.49]	0.36 [4.28]	0.15 [1.91]	0.22 [3.01]	0.11 [1.18]
Anomaly 1	0.19 [8.72]						0.90 [1.32]
Anomaly 2		0.17 [9.20]					-0.80 [-1.66]
Anomaly 3			0.19 [6.35]				0.10 [1.80]
Anomaly 4				0.41 [6.95]			0.20 [0.34]
Anomaly 5					0.10 [8.29]		0.50 [2.33]
Anomaly 6						0.17 [7.58]	0.88 [3.14]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCM} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [2.09]	0.15 [2.10]	0.14 [2.03]	0.16 [2.32]	0.16 [2.26]	0.16 [2.27]	0.14 [2.01]
Anomaly 1	18.72 [4.99]						15.86 [3.08]
Anomaly 2		10.84 [2.75]					-0.78 [-0.15]
Anomaly 3			14.29 [4.13]				12.05 [3.19]
Anomaly 4				6.09 [2.25]			6.29 [2.22]
Anomaly 5					-0.72 [-0.18]		-6.87 [-1.64]
Anomaly 6						4.05 [1.29]	-1.73 [-0.52]
mkt	-5.17 [-3.28]	-4.96 [-3.10]	-3.24 [-1.96]	-5.32 [-3.32]	-5.22 [-3.24]	-5.27 [-3.28]	-3.62 [-2.18]
smb	4.42 [1.83]	4.96 [2.02]	10.66 [4.03]	6.74 [2.76]	6.13 [2.50]	6.09 [2.50]	9.80 [3.51]
hml	-10.19 [-3.35]	-10.22 [-3.30]	-9.36 [-3.03]	-11.13 [-3.60]	-10.88 [-3.50]	-11.37 [-3.64]	-8.73 [-2.82]
rmw	1.07 [0.34]	2.16 [0.68]	-5.41 [-1.45]	4.01 [1.27]	3.13 [0.99]	3.21 [1.02]	-4.89 [-1.29]
cma	20.29 [4.30]	21.51 [4.42]	15.84 [3.07]	19.73 [3.74]	25.99 [3.82]	21.92 [4.12]	17.56 [2.52]
umd	2.23 [1.41]	2.51 [1.54]	3.34 [2.12]	2.82 [1.74]	3.44 [2.15]	3.44 [2.15]	1.61 [0.99]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	11	12	10	10	10	15

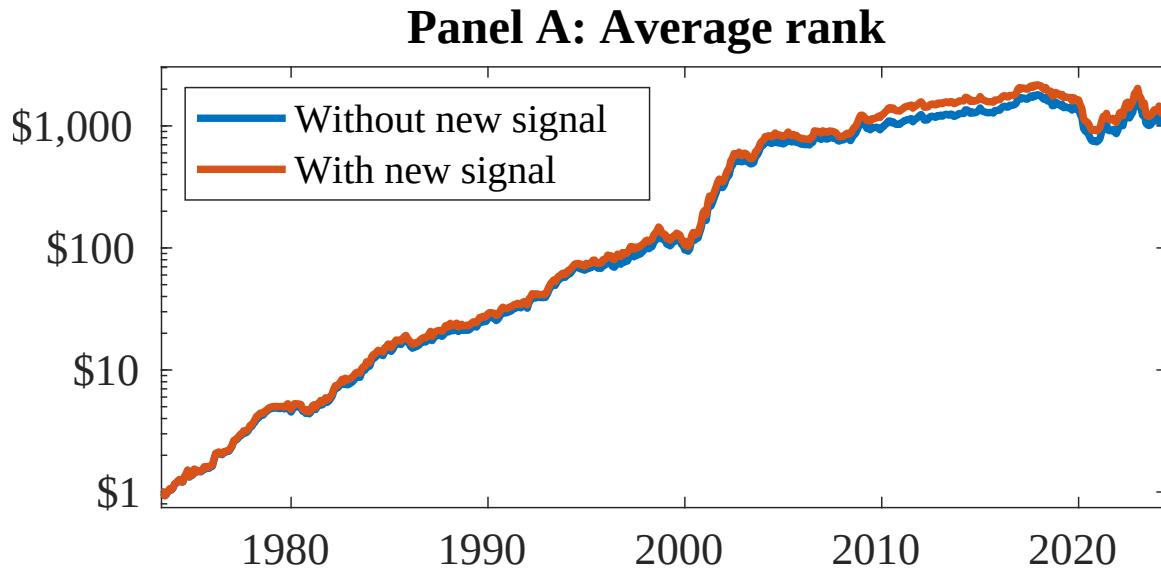


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCM. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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